

Portfolio Exam 3

Reinforcement Learning – BlackJack Player

Due date: 10.07.2026, 23:59

This portfolio exam is about a self-learning BlackJack player based on Reinforcement Learning methods. The foundations of Reinforcement Learning are described in [2]. Basic concepts of the BlackJack card game, various rule variations, and card counting methods can be found in [3].

Upload your personal deliverable via the eLearning platform. The code should be structured in such a way that it can be executed without modification. So please include a readme file.

Task P3.1

Realize a Reinforcement Learning implementation of a self-learning BlackJack player in a programming language of your choice. This implementation provides the basis for the paper from Task P3.2. It shall learn optimal policies for at least the following scenarios:

1. The “*Basic Strategy*” from [3].
2. The “*Complete Point-Count System*” from [3] including an adjustment of the player’s bet.
3. In addition to the basic rules, two rule variations of your choice shall be examined for their influence on the strategies from (1.) and (2.).
4. Consider improving the system from (2.) to be able to achieve higher profits on average. Note: Your system does not need to be suitable for humans. It may therefore be relatively complicated, e.g., with respect to card counting.

What profit can be expected for the different scenarios in a greedy evaluation of your approaches?

The deliverable for this task is the commented source code of your implementation and all logfiles that contributed to the results in Task P3.2. Do not use an external Reinforcement Learning framework.

Task P3.2

Prepare a research paper using the official IEEE conference template from [1] (format A4). Use either the L^AT_EX or the Overleaf template. Do not use Microsoft Word with the IEEE Word template for the paper. The maximum length of the paper is 6 pages.

Make sure your paper has an appropriate structure and outline. If you are unsure how well good papers are structured, read high-quality papers and analyze their structure. An appropriate structure might look like this:

- *Abstract*: One-paragraph summary of the paper. The Abstract provides a short overview of the paper.
- *Introduction*: What is the topic and why is it worth studying? The Introduction commonly describes the topic under investigation, summarizes or discusses relevant prior research, identifies open questions and problems and provides an overview of the research that is to be described in greater detail in the sections to follow.
- *Description of Reinforcement Learning player*: What did you do? This is a section which details how the work was performed. It typically features a description of the methods that were involved. As a rule of thumb, this section should be sufficiently detailed for another researcher to reproduce your work. You should also address the theoretical background of your specific problem solutions for the task at hand, taking into account the issues raised above.

Portfolio Exam 3

Reasoning and Decision
Making under Uncertainty
Summer 2026

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Technical University of Applied Sciences
Würzburg-Schweinfurt

- *Experiments and Evaluation:* How well does it work? The evaluation must show that your implementation works correctly. Use appropriate datasets to represent specific properties of the algorithm and discuss them.
- *Conclusion:* A brief summary and outlook of questions to be explored in the future.
- *References:* List of articles, books, etc. cited. A list of the sources that are cited in the paper.

The practical part from Task P3.1 focuses on different aspects from the application point of view. The paper should integrate these findings into the paper. Among other things, the following scientific questions arise:

- How do the different learning algorithms behave for the task at hand? Which learning methods have specific advantages and disadvantages here?
- Can you estimate the size of the state-action space for your implementation? Can one expect to achieve stable estimates $Q(\cdot, \cdot)$? If not, how do you deal with this?
- Can you explain why the rule changes you decided to make led to the policy changes you observed?
- Can you show that your learned profit estimates are correct?

The deliverable for this task is a pdf version of your paper.

References

- [1] IEEE Paper Template. <https://www.ieee.org/conferences/publishing/templates.html>.
- [2] Richard S. Sutton and Andrew G. Barto. *Reinforcement Learning: An Introduction*. MIT Press, Cambridge, MA, 2 edition, 2018.
- [3] Edward O. Thorp. *Beat the Dealer*. Vintage, New York, 1966.